

Emotion Recognition System Based on Voice and Facial Expressions

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Abstract

In recent years, advances in emerging technologies, such as facial recognition and voice detection, have significantly improved our daily lives. In the DSP Lab course throughout this semester, we utilized tools including Python, MATLAB, and Arduino to implement various applications, ranging from image processing to heart rate monitoring. Using the knowledge acquired in this course, we aim to develop a system for human emotion recognition. Since emotions are typically manifested through facial muscle movements and vocal characteristics, our system determines the subject's current emotional state by analyzing dynamic changes in facial expressions and acoustic waveform patterns.

1 Introduction

Our project proposes a multi-modal emotion recognition system that integrates visual facial expressions and acoustic features. Unlike traditional deep learning approaches, our system employs a baseline-calibration mechanism to address individual differences in facial structure and vocal characteristics. We divide emotion recognition into visual module and audio module. In this way, we can make recognition more precise by using two different ways to recognize the emotion. In this project, the system is designed to successfully distinguish 4 main emotions among human beings, which are "neutral, happiness, sadness, and anger."

2 Proposed Method

The system operates on a synchronized state-machine architecture comprising a calibration phase and a recognition phase. To mitigate the bias caused by a subject's unique facial structure or natural voice pitch, we introduce a Relative Calibration mechanism.

During the calibration phase ($t \in [0, 5]$ seconds), the system continuously captures visual frames and audio samples to compute a personalized baseline. Instead of averaging output probabilities, we calculate specific feature baselines:

- **Visual Baseline ($\mathbf{B}_{geo}$):** The average Brow-Eye Ratio and Mouth Frown Ratio calculated from the MediaPipe Face Mesh.
- **Audio Baseline ($\mathbf{B}_{audio}$):** The average vector of extracted acoustic features (including Pitch, Energy, and Spectral Centroid).

The subsequent emotion recognition is performed by analyzing the relative deviations (ratios or deltas) of the current frame's features from these personalized baselines.

2.1 Visual Module

The visual module utilizes a hybrid approach, combining the texture analysis of the DeepFace framework with the precise geometric tracking of MediaPipe Face Mesh. To ensure robustness against camera distance, all geometric measurements are normalized by the Face Height (H_{face}).

2.1.1 Motivation and Design Rationale

Conventional Convolutional Neural Networks (CNNs), such as DeepFace, excel at texture-based feature extraction, but often struggle with two fundamental challenges in real-world emotion recognition:

1. **Variability of Human’s Facial Expression**

Fixed geometric thresholds fail to account for natural variations in facial structure, leading to persistent bias in static analysis.

2. **Ambiguity among Different Emotions**

For example, high-arousal negative emotions (e.g. intense anger) and low-arousal negative emotions (e.g. sadness) share similar lower-face morphological features. Texture-based classifiers frequently misclassify these emotions due to these shared dental features.

To address these limitations, we propose a Relative Geometric Calibration framework integrated with Hierarchical Decision Logic. This approach shifts the judging methods from absolute shape recognition to relative dynamic analysis, enforcing physiological constraints to resolve ambiguities.

2.1.2 Geometric Feature Definition

The system utilizes MediaPipe Face Mesh to extract 468 3D facial landmarks. To ensure scale invariance regarding the distance between the subject and the camera, all geometric measurements are normalized by the Face Height (H_{face}), defined as the Euclidean distance between the hairline landmark (P_{10}) and the chin landmark (P_{152}):

$$H_{face} = |y_{152} - y_{10}| \quad (1)$$

Based on this normalization, we define two critical geometric descriptors to capture emotional dynamics:

- **Brow-Eye Compression Ratio (R_{brow})**

This metric quantifies the vertical proximity of the eyebrows to the eyes, serving as the primary indicator for anger.

$$R_{brow} = \frac{|y_{159} - y_{52}|}{H_{face}} \quad (2)$$

where P_{52} represents the left inner brow and P_{159} represents the upper margin of the left eye.

- **Mouth Frown Ratio (R_{frown})**

This metric captures the vertical displacement of the mouth corners relative to the lips’ center to distinguish between sadness and happiness.

$$R_{frown} = \frac{\frac{y_{61} + y_{291}}{2} - \frac{y_{13} + y_{14}}{2}}{H_{face}} \quad (3)$$

where P_{61}, P_{291} are the mouth corners, and P_{13}, P_{14} are the upper and lower lip centers. A positive value implies a downturned mouth (sadness), while a negative value implies an upturned mouth (happiness).

2.1.3 Relative Calibration Mechanism

Instead of using fixed absolute thresholds, the system employs a Relative Calibration phase. During the first 5 seconds, the system computes the temporal average of the geometric features to establish a personalized baseline vector $\mathbf{B}_{geo} = \{\mu_{brow}, \mu_{frown}\}$.

During the inference phase, the system computes the relative deviation of the current frame t from this baseline. We employ two distinct deviation metrics tailored to the biomechanics of each feature:

- **Brow Change Percentage:** $\Delta R_{brow}^{(t)} = \frac{R_{brow}^{(t)}}{\mu_{brow}}$
- **Mouth Deviation:** $\Delta R_{frown}^{(t)} = R_{frown}^{(t)} - \mu_{frown}$

2.1.4 Hierarchical Decision Logic

To resolve the remained confusion cases among the 4 emotions, we implemented a hierarchical decision tree with an Absolute Veto mechanism:

1. Angry Priority Check

The system first checks if the brow distance has dropped significantly. If the brow distance is less than 88% of the baseline ($\Delta R_{brow}^{(t)} < 0.88$), the Anger score is boosted by a gain factor of 15 proportional to the intensity:

$$S_{angry} \leftarrow S_{angry} + \min((0.88 - \Delta R_{brow}^{(t)}) \times 15, 0.9) \quad (4)$$

and the happiness score is forcibly set to 0:

$$S_{happy} \leftarrow 0.0 \quad (5)$$

This effectively eliminates false positives caused by dental texture. Also, for Neutral and Sadness scores, they are suppressed to ensure the dominance of the detected high-arousal state:

$$S_{sad} \leftarrow S_{sad} - 0.3 \times \min((0.88 - \Delta R_{brow}^{(t)}) \times 15, 0.9) \quad (6)$$

$$S_{neutral} \leftarrow \max(0, S_{neutral} - 2.0 \times \min((0.88 - \Delta R_{brow}^{(t)}) \times 15, 0.9)) \quad (7)$$

2. Secondary Happy Check

If the anger condition is not met, the system evaluates the mouth curvature to distinguish between happiness and sadness. If the mouth corners rise significantly relative to the baseline ($\Delta R_{frown}^{(t)} < -0.015$), the Happiness score is boosted by a gain factor of 15 based on the magnitude of the rise, capped at 0.8. Conflicting emotions scores (Sad and Angry) are suppressed by being set to zero:

$$S_{happy} \leftarrow S_{happy} + \min(\Delta R_{frown}^{(t)} \times 15, 0.8) \quad (8)$$

$$S_{angry} \leftarrow 0.0 \quad (9)$$

$$S_{sad} \leftarrow 0.0 \quad (10)$$

$$S_{neutral} \leftarrow \max(0, S_{neutral} - 1.5 \times \min(\Delta R_{frown}^{(t)} \times 15, 0.8)) \quad (11)$$

3. Final Sad Check

If the mouth corners drop relative to the baseline ($\Delta R_{frown}^{(t)} > 0.015$), the Sadness score is boosted by a gain factor of 20. A higher gain is used here to amplify the typically subtle visual cues of sadness:

$$S_{sad} \leftarrow S_{sad} + \min((\Delta R_{frown}^{(t)} - 0.015) \times 20, 0.6) \quad (12)$$

$$S_{angry} \leftarrow S_{angry} - 0.2 \quad (13)$$

$$S_{happy} \leftarrow 0.0 \quad (14)$$

$$S_{neutral} \leftarrow \max(0, S_{neutral} - 1.5 \times \min((\Delta R_{frown}^{(t)} - 0.015) \times 20, 0.6)) \quad (15)$$

2.1.5 Temporal Smoothing

To ensure output stability and filter out jitter from frame-to-frame model inference, an Exponential Moving Average (EMA) filter is applied to the final probability scores:

$$S_{final}^{(t)} = \alpha \cdot S_{corrected}^{(t)} + (1 - \alpha) \cdot S_{final}^{(t-1)} \quad (16)$$

We utilize a smoothing factor of $\alpha = 0.6$, empirically selected to balance responsiveness to micro-expressions with the suppression of noise.

2.2 Audio Module

2.2.1 Motivation and Design Rationale

In speech emotion recognition, a pervasive challenge is the variability in vocal characteristics. For instance, a speaker with a naturally high-pitched voice might be misclassified as "excited" or "angry" by a model trained on lower-pitched speakers. Similarly, the "loudness" of speech is highly dependent on the microphone distance rather than the emotional state.

To address this, we propose a Relative Acoustic Calibration framework. Instead of analyzing absolute spectral values, our system computes the deviation of the current utterance from a 5-second neutral baseline. Furthermore, to ensure robustness without a massive labeled dataset, we employ a Psychoacoustic Heuristic Classifier rather than a black-box neural network. This allows us to explicitly codify acoustical fingerprints into the decision logic.

2.2.2 Acoustic Feature Extraction

We extract a comprehensive set of 13 acoustic features ($f \in \mathbb{R}^{13}$) using the Librosa library, categorized into three domains:

1. Prosodic Features:

- **Pitch (F_0):** Extracted mean fundamental frequency using the probabilistic YIN algorithm (pYIN).
- **Pitch Variance (σ_{F_0}):** Standard deviation of F_0 , indicating intonation range.
- **Energy (E_{RMS}):** P80 (80th percentile) Root-Mean-Square energy.
- **Energy Variance (σ_E):** Calculated for both Pitch and Energy to measure signal dynamics.

2. Spectral Features:

- **Spectral Centroid (SC):** Center of mass of the spectrum, represents the "brightness" of the sound.
- **Spectral Flux (SF):** Measures the rate of change in the power spectrum, indicating temporal chaos.
- **High Frequency Energy (E_{hi}):** The sum of energy > 1500 Hz.
- **Low Frequency Energy (E_{lo}):** The sum of energy < 1500 Hz.
- **MFCC Std:** Standard deviation of Mel-frequency cepstral coefficients.
- **Delta Std:** Standard deviation of the first-order derivative of MFCCs.

3. Voice Quality Features:

- **Harmonics-to-Noise Ratio (HNR):** Measures voice purity. Low HNR indicates hoarseness or breath.
- **Zero-Crossing Rate (ZCR):** Correlates with fricatives and noisy excitation.
- **Spectral Slope:** The slope of the MFCC trajectory.

2.2.3 Relative Calibration Mechanism

During the calibration phase ($t \in [0, 5]$), the system records the subject's neutral voice to compute a baseline vector $\mathbf{B}_{audio}$. For every subsequent inference window, we compute the Relative Feature Ratio (r_i):

$$r_i = \frac{f_i^{(current)}}{f_i^{(baseline)}} \quad (17)$$

A ratio $r_i > 1.0$ indicates an increase in the feature value relative to the neutral state.

2.2.4 Heuristic Classification Logic

The classification is performed via a hierarchical rule-based inference engine. r_i is the relative ratio of feature i compared to the baseline, $r_i = \frac{f_i^{(current)}}{f_i^{(baseline)}}$.

1. Arousal and Valence Detection

We define binary indicator functions $\mathbb{I}(\cdot)$ to detect high arousal and positive valence states.

- **Arousal Detection (Intensity)**

The Arousal score $S_{arousal}$ is determined by significant increases in Pitch(F_0), Energy(E_{RMS}), and Flux(SF); Energy is given a higher weight of 1.5:

$$S_{arousal} = \mathbb{I}(r_{F_0} > 1.15) + 1.5 \cdot \mathbb{I}(r_E > 1.20) + \mathbb{I}(r_{SF} > 1.5) \quad (18)$$

If $Score_{Arousal} \geq 1.5$, the state is HIGH AROUSAL, else LOW AROUSAL.

- **Valence Detection (Positivity)**

Similarly, Valence is assessed using brightness and spectral harmonicity. A positive valence score $S_{valence}$ accumulates evidence from Spectral Centroid(SC), Harmonics-to-Noise Ratio (HNR), and Spectral Slope:

$$\begin{aligned} Score_{Valence} = & \mathbb{I}(r_{SC} > 1.05) + \mathbb{I}\left(\frac{E_{hi}}{E_{lo}} > 0.90\right) + \mathbb{I}(r_{HNR} > 0.85) \\ & + \mathbb{I}(r_{SF} < 3.5) + \mathbb{I}(r_{SMFCC} > 1.00) \\ & + 1.5 \cdot \mathbb{I}(r_{\sigma_{F_0}} > 1.2 \text{ and } r_{F_0} > 1.1) \end{aligned} \quad (19)$$

If $Score_{Valence} \geq 3.0$, the state is POSITIVE, else NEGATIVE.

2. Categorical Emotion Scoring

We accumulate scores for each emotion based on their distinct acoustic fingerprints defined in the code.

- **Happy / Laughter (S_{laugh})**

Characterized by brightness, moderate chaos, and high pitch variance ("bounciness"):

$$\begin{aligned} S_{laugh} = & \mathbb{I}(r_{SC} > 1.10) + \mathbb{I}(2.0 < r_{SF} < 8.0) \\ & + \mathbb{I}(r_{HNR} < 0.75) + \mathbb{I}(r_{ZCR} > 1.2) \\ & + 2.0 \cdot \mathbb{I}(r_{\sigma_{F_0}} > 1.5) \quad (\text{Strong Bounciness}) \\ & + \mathbb{I}(r_{F_0} > 1.2 \text{ and Valence is POS}) \end{aligned} \quad (20)$$

- **Sadness (S_{sad})**

Defined by suppressed dynamics (monotonicity) and low energy:

$$S_{sad} = \mathbb{I}(r_{\sigma_{F_0}} < 0.85) + \mathbb{I}(r_E < 0.8) + \mathbb{I}(r_{\sigma_E} < 0.85) + \mathbb{I}(r_{ZCR} < 0.85) \quad (21)$$

- **Crying (S_{cry})**

A high-arousal variant of sadness, distinguished by extreme pitch variance but lower spectral brightness:

$$\begin{aligned} S_{cry} = & 2.0 \cdot \mathbb{I}(r_{\sigma_{F_0}} > 2.0) + 1.5 \cdot \mathbb{I}(r_{SC} < 0.95) \\ & + \mathbb{I}(1.5 < r_{SF} < 5.0) - \mathbb{I}(r_{SC} > 1.15) \end{aligned}$$

- **Angry (S_{angry})**

Characterized by roughness, chaos, and loudness. Negative constraints are applied to differentiate from laughter or happy screams:

$$\begin{aligned}
S_{angry} = & \mathbb{I}(r_{HNR} < 0.65) + \mathbb{I}(r_{SF} > 4.5) \\
& + \mathbb{I}(r_{ZCR} > 1.6) + \mathbb{I}(r_{\sigma_E} > 3.0) \\
& + \mathbb{I}(r_E > 1.6) + \mathbb{I}(r_{SC} > 1.1) \\
& - 2.0 \cdot \mathbb{I}(r_{\sigma_{F0}} > 1.5) \quad (\text{Too Bouncy}) \\
& - 2.0 \cdot \mathbb{I}(r_{F0} > 1.3 \text{ and Valence is POS})
\end{aligned} \tag{22}$$

3. Final Decision Strategy

The final audio prediction follows a competitive logic:

- **If Arousal is HIGH**

The winner is selected from $\{Angry, Cry, Happy\}$ based on the maximum score.

- **Tie-Breaker (Valence)**

If $|S_{angry} - S_{laugh}| \leq 2.0$ and Valence is POSITIVE, the result is shifted to Happy.

- **Tie-Breaker (Timbre)**

If the winner is Angry or Sad (Cry) and scores are close, r_{SC} is used: darker sound ($r_{SC} < 1.0$) implies Sad/Cry, while brighter sound implies Angry.

- **Confidence Threshold:** If the winning score < 1.0 , output Neutral.

- **If Arousal is LOW**

- **If $S_{sad} \geq 2$:** output Sad.

- **Otherwise:** output Neutral.

2.3 Multimodal Fusion and Decision Strategy

2.3.1 Weighted Decision-Level Fusion

To integrate the disparate signals from the visual and audio modules, we employ a Weighted Late Fusion strategy. This approach allows the system to process modalities asynchronously and combine them at the scoring level.

Let V be the visual score vector derived from geometric deltas, and A be the normalized acoustic score vector. The fused score S_{fused} for each emotion category $e \in \{Angry, Sad, Happy, Neutral\}$ is computed as a linear combination:

$$S_{fused}[e] = w_{vis} \cdot V[e] + w_{audio} \cdot A[e] \tag{23}$$

where the modality weights are set equally to $w_{vis} = 0.5$ and $w_{audio} = 0.5$.

This balanced weighting assumes that facial expressions and vocal intonations contribute equally to the emotional percept in a general interaction context.

2.3.2 Temperature-Scaled Softmax Normalization

A critical innovation in our decision logic is the treatment of the "Neutral" state. In many traditional systems, Neutral is defined passively (i.e., "if no emotion score exceeds threshold τ , then Neutral"). However, this approach often leads to unstable flipping between low-confidence states. Instead, we treat Neutral as an active class that competes directly with other emotions. To convert the

raw fused scores into a normalized probability distribution, we apply a Temperature-Scaled Softmax function:

$$P(e_i) = \frac{\exp(S_{fused}[e_i]/T)}{\sum_{j \in Emotions} \exp(S_{fused}[e_j]/T)} \quad (24)$$

Design Rationale for Softmax:

1. Probabilistic Interpretation

Softmax ensures that the output values sum to 1.0, providing an interpretable confidence metric $P(e)$.

2. Competition Mechanism

By including Neutral in the denominator, high scores in active emotions (e.g., Anger) naturally suppress the Neutral probability, while low scores across all active emotions naturally result in a high Neutral probability.

3. Temperature Scaling (T)

We utilize a temperature parameter $T = 0.8$. A temperature $T < 1.0$ "sharpens" the distribution, making the model more decisive by amplifying the probability of the leading class. This reduces ambiguity in the final output.

2.3.3 Final Decision

The final predicted emotion E^* is determined by selecting the class with the maximum probability:

$$E^* = \underset{e \in \{Angry, Sad, Happy, Neutral\}}{\arg \max} P(e) \quad (25)$$

This methodology ensures that the system defaults to "Neutral" smoothly when emotional signals are weak, without requiring an arbitrary manual threshold cut-off.

3 Results and Discussion

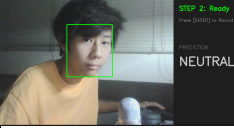
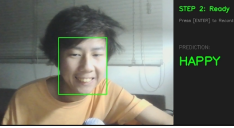
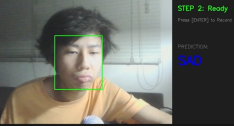
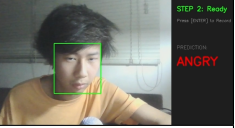
Emotions	Neutral	Happy	Sad	Angry
Visual Result				
Score Result (Fused Score)	angry : 0.0 sad : 0.16666 happy : 0.02140 neutral : 0.33333 WINNER : NEUTRAL	angry : 0.03968 sad : 0.45 happy : 0.68175 neutral : 0.04999 WINNER : HAPPY	angry : 0.02701 sad : 0.40111 happy : 0.0 neutral : 0.33333 WINNER : SAD	angry : 0.55877 sad : 0.33333 happy : 0.2 neutral : 0.09999 WINNER : ANGRY

Table 1: Performance Comparison of the System Across Neutral, Happy, Sad, and Angry Emotional States

Table 1 presents the performance evaluation of the proposed emotion recognition system across four distinct emotional states: Neutral, Happy, Sad, and Angry. The qualitative results (visual bounding boxes) and quantitative results (fused scores) demonstrate the system’s capability to correctly identify the subject’s emotional state in real-time scenarios.

As shown in the score breakdown, the system successfully classified the “Winner” emotion consistent with the ground truth in all test cases. For high-arousal emotions, the distinction was particularly evident. In the **Happy** state, the system yielded a dominant fused score of 0.68175, significantly surpassing the second-highest score (Sad: 0.45). Similarly, the **Angry** state was correctly identified with

a score of 0.55877.

For low-arousal emotions, which are typically harder to distinguish, the system maintained robustness. The **Sad** state achieved a winning score of 0.40111, successfully differentiating it from the Neutral state (0.33333), despite the subtle facial feature differences. The **Neutral** state was also accurately detected with the highest probability among candidates.

4 Conclusion

This study implemented and validated a multimodal emotion recognition system. The experimental results confirm that the proposed scoring mechanism effectively integrates feature data to predict emotional states with high accuracy. The system exhibited strong performance not only in distinct, high-intensity emotions like Happiness and Anger but also showed capability in distinguishing subtle variations between Sadness and Neutral states. The consistency between the visual detection results and the calculated fused scores validates the reliability of our approach for real-world applications.

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